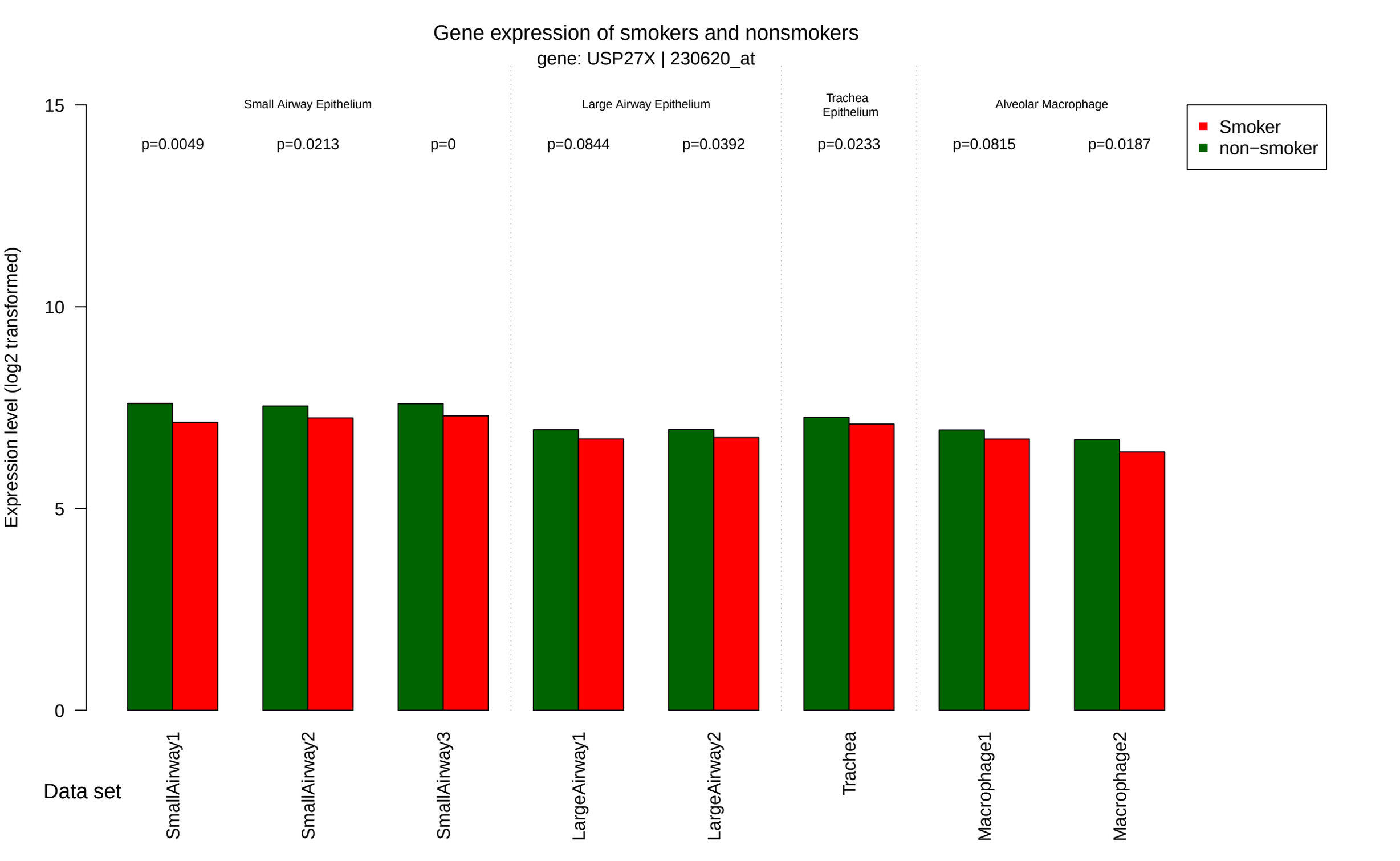
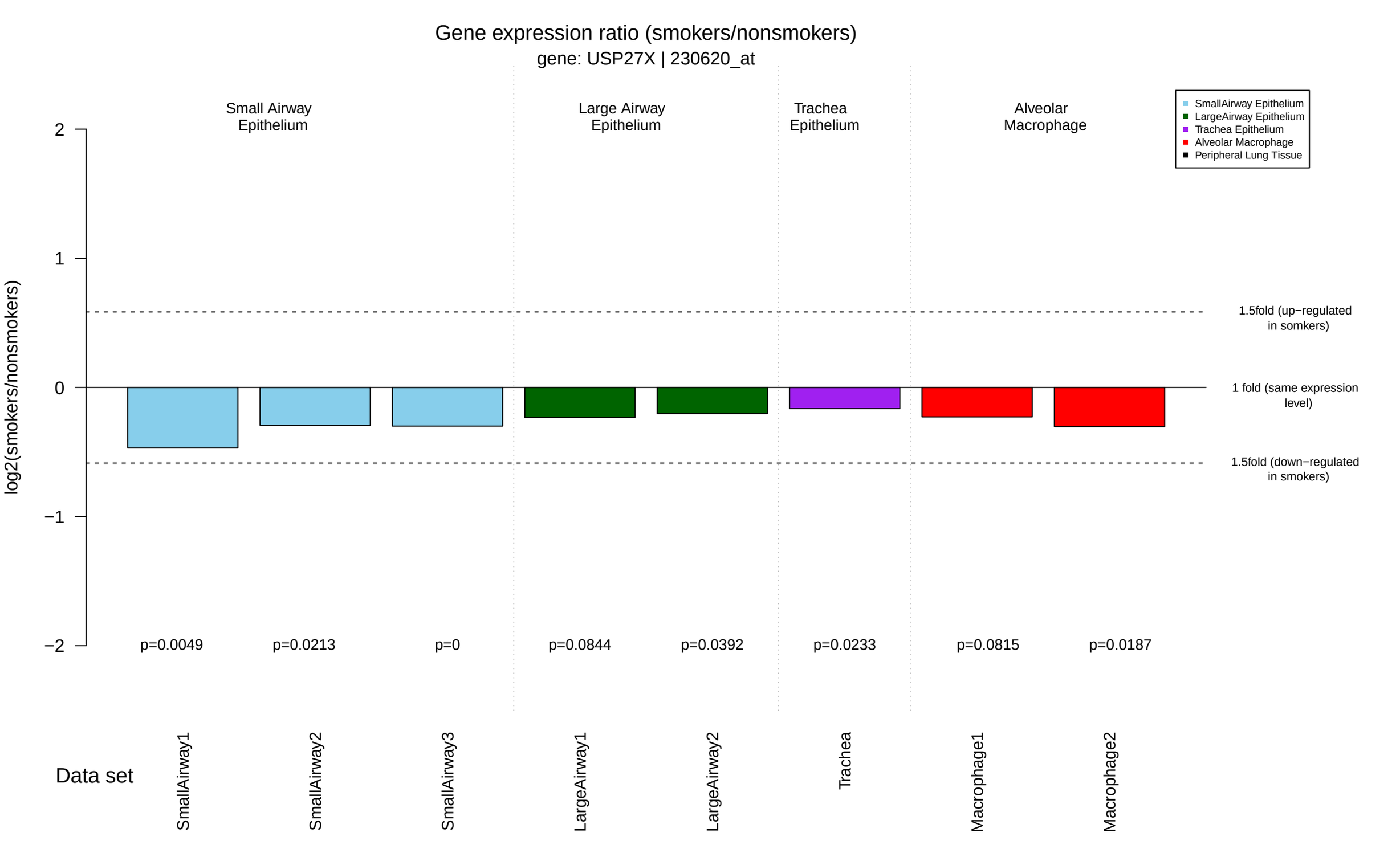
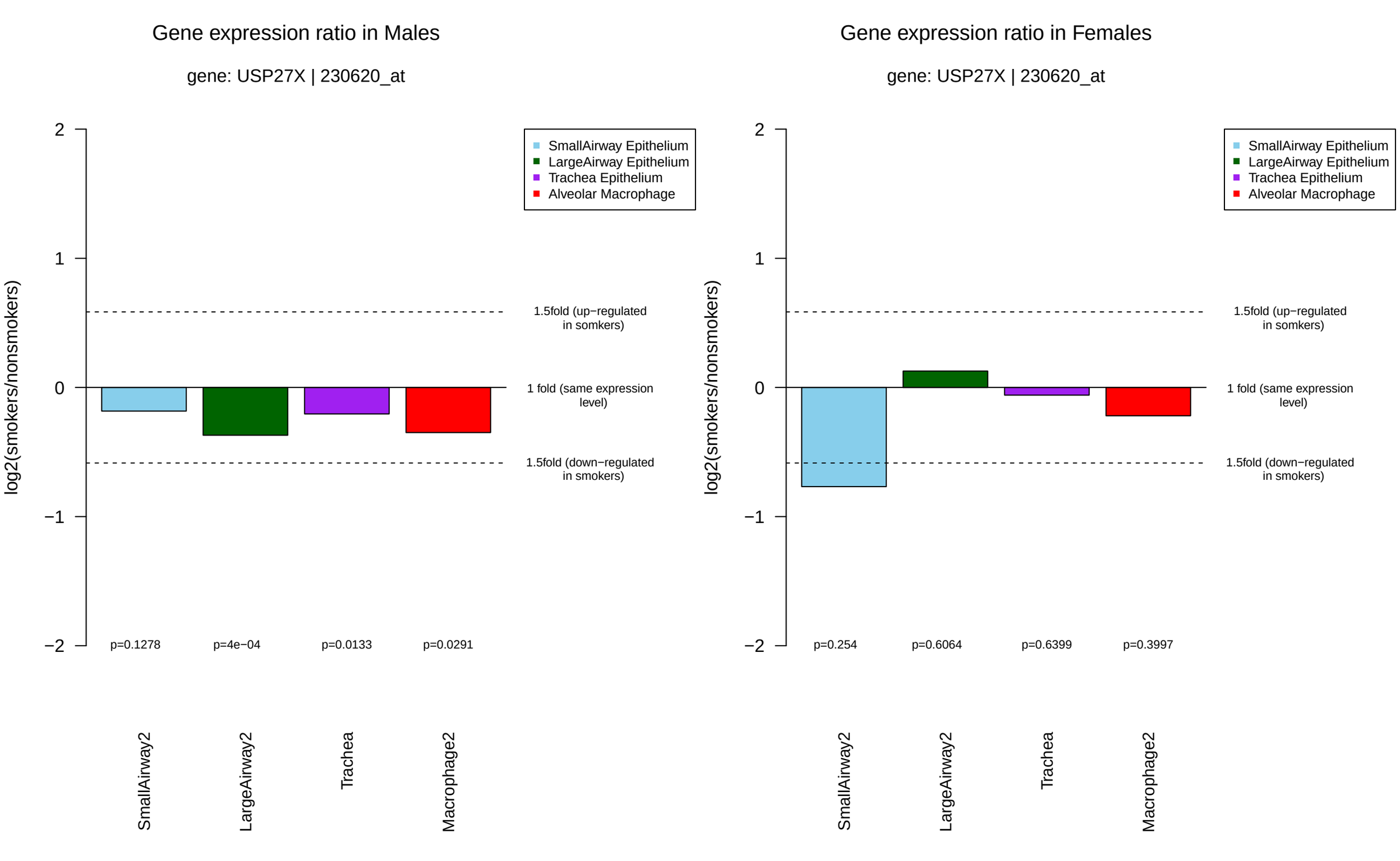
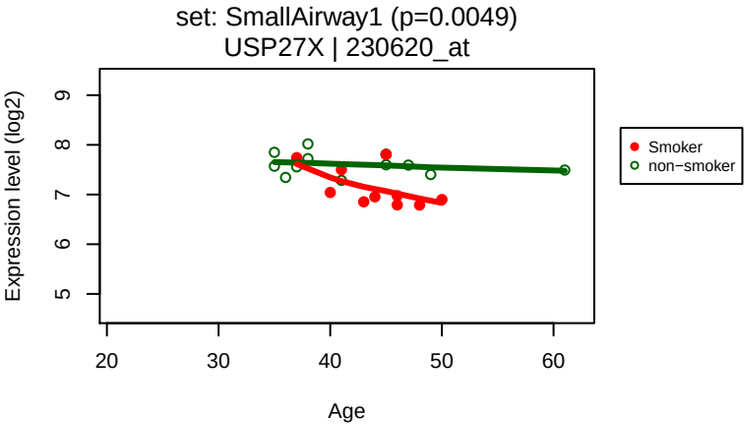
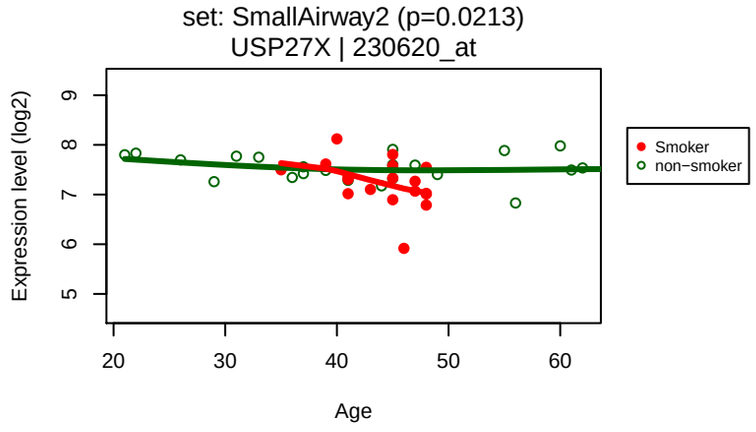
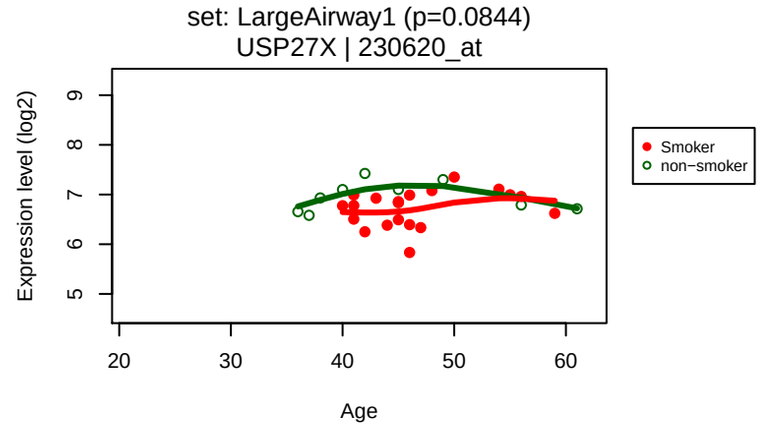

Gene expression of smokers and nonsmokers
gene: USP27X | 230620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0049
p=0.0213
p=0
p=0.0844
p=0.0392
p=0.0233
p=0.0815
p=0.0187
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

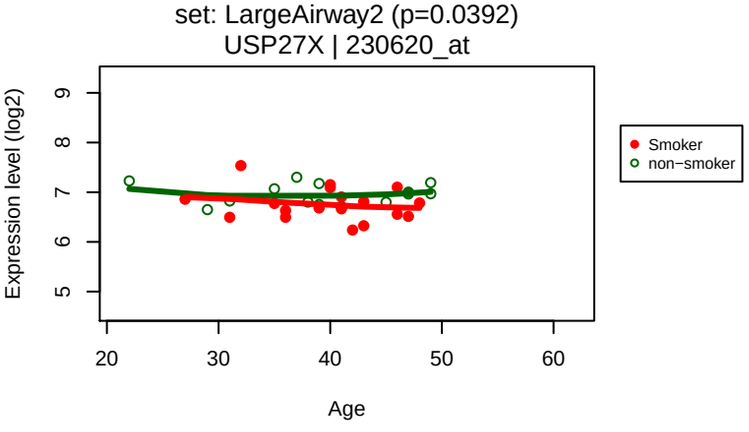

Gene expression ratio (smokers/nonsmokers)
gene: USP27X | 230620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=0.0213
p=0
p=0.0844
p=0.0392
p=0.0233
p=0.0815
p=0.0187
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

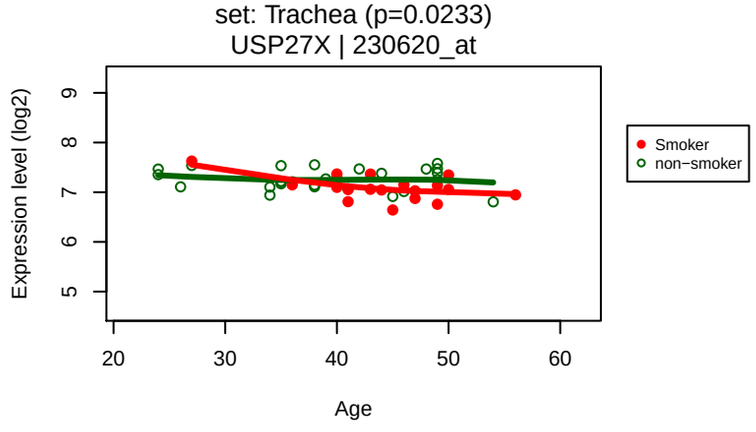

Gene expression ratio in Males
gene: USP27X | 230620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1278
p=4e−04
p=0.0133
p=0.0291
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP27X | 230620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.254
p=0.6064
p=0.6399
p=0.3997
SmallAirway2
LargeAirway2
Trachea
Macrophage2

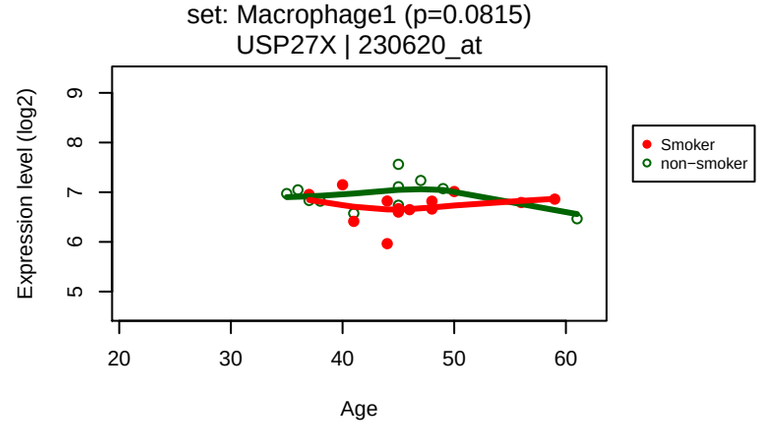

set: SmallAirway1 (p=0.0049)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker

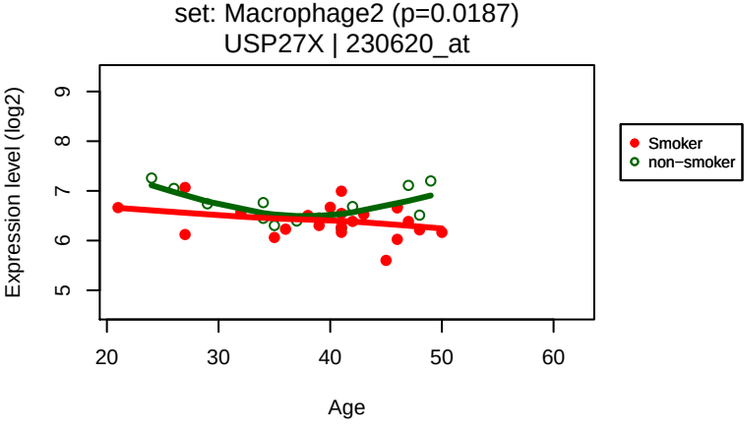

set: SmallAirway2 (p=0.0213)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker

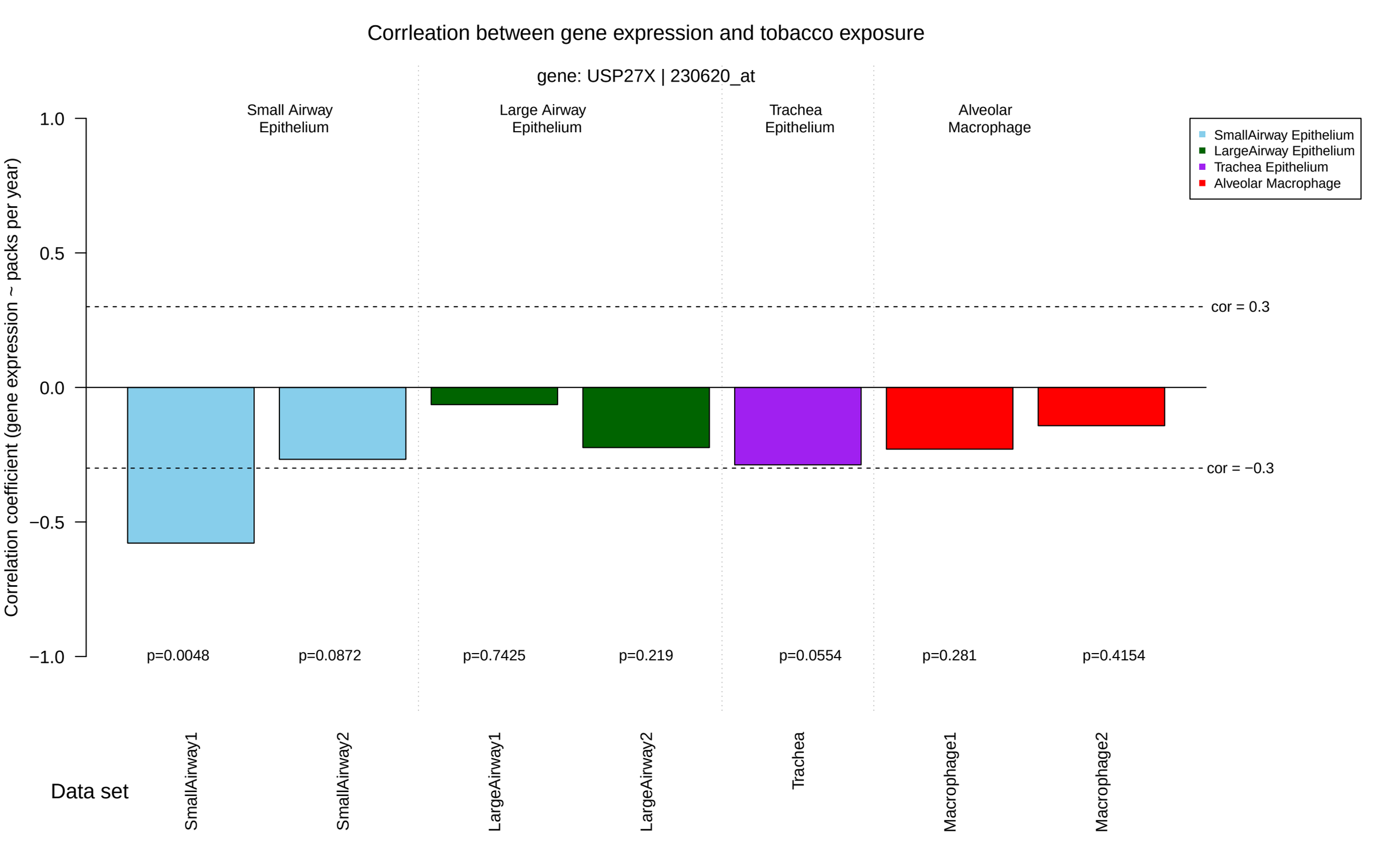

set: LargeAirway1 (p=0.0844)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0392)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0233)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0815)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0187)
USP27X | 230620_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP27X | 230620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0048
p=0.0872
p=0.7425
p=0.219
p=0.0554
p=0.281
p=0.4154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2